



UNIVERSITY OF MAKATI

J.P. Rizal Extension, West Rembo Makati City



COLLEGE OF COMPUTING AND INFORMATION SCIENCES

October 6, 2024

Asst. Prof. Ali A. Naim, MIS
Program Coordinator
Computer Science Program (OIC), IS and ITE Allied Program

Prof. Daniel P. Dellosa
Thesis 1 Course Adviser
College of Computing and Information Sciences

Dear Sir:

Good day!

Clyde Canlas, Jhon Mark Enrique, Patrick Formanes, John Elvin Recto, bona fide fourth-year **Bachelor of Science in Computer Science Major in Application Development** students, are writing to formally request your approval to change the title of their thesis from **"DEVELOPMENT OF VERIFYID: AN ONLINE BARANGAY CERTIFICATION SYSTEM WITH IDENTITY AUTHENTICATION THROUGH INTEGRATION OF IMAGE RECOGNITION"** to **"DEVELOPMENT OF CERTIFACE: ONLINE BARANGAY SYSTEM WITH IDENTITY AUTHENTICATION USING FACE RECOGNITION FOR BARANGAY 781 ZONE 85 STA. ANA MANILA.**

The primary reasons for this proposed change are:

1. The technical adviser recommends including all the barangay operations not only exclusive to requesting certificates or documents.
2. The technical adviser recommends including the name of the barangay in the title.

A brief summary detailing the changes in the scope and focus of the thesis, which justify this title change, is attached. The proponents believe that the new title will enhance the clarity and impact of the thesis. We respectfully seek your approval for this modification and are available to discuss this further or provide any additional information if needed.

We are hoping for your favorable action on this matter. Thank you for your time and attention.

Very truly yours,

Clyde D. Canlas
Student

Jhon Mark A. Enrique
Student

Patrick Formanes
Student

John Elvin P. Recto
Student

Approved By:

Asst. Prof. Era Marie F. Gannaban
Panel Chair

Asst. Prof. Mary Ellaine R. Cervantes, MIS, MTE
Panel 2

Assoc. Prof. Lester G. Diampoc
Panel 3

Noted by:

Asst. Prof. Alexander B. Pahayahay
Technical Adviser

Asst. Prof. Daniel P. Dellosa
Course Adviser

Asst. Prof. Ali A. Naim, MIS
Dept. Chair, Computer Science Programs

K-Nearest Neighbors Performance:

Cross-validation scores: [0.55555556 0.66666667 0.75 0.625 0.75]
Mean CV Accuracy: 0.6694 (+/- 0.1495)
Precision: 0.7863
Recall: 0.6667
F1 Score: 0.5865

Multinomial Naive Bayes Performance:

Cross-validation scores: [0.77777778 0.88888889 0.75 1. 0.75]
Mean CV Accuracy: 0.8333 (+/- 0.1956)
Precision: 1.0000
Recall: 1.0000
F1 Score: 1.0000

Logistic Regression Performance:

Cross-validation scores: [0.55555556 0.88888889 0.75 1. 0.75]
Mean CV Accuracy: 0.7889 (+/- 0.2994)
Precision: 0.9179
Recall: 0.9048
F1 Score: 0.9019

Top 10 important features:

tanga: 0.3618
ka: 0.3618
luka: 0.2811
unggoy: 0.2316
satanas: 0.2316
pangit: 0.2316
isinusumpa: 0.2316
demonyo: 0.2316
bastardo: 0.2316
bakla: 0.2316

Model Comparison:

KNN: A non-parametric method, doesn't make assumptions about data distribution.
Multinomial Naive Bayes: Assumes feature independence, works well with text data.
Logistic Regression: Models probability of classes, can handle feature interactions.